

Steps:

Dataset Selection (NCBI Nucleotide Search, Webversion)

1. When you have selected your species and gene, search *NCBI Nucleotide Search* and open the link that says *Home - Nucleotide - NCBI - NIH*
2. To find the sequences you want, search “*gene genus*”
 - a. EX: rbcL *Vaucheria*
3. Once you find the sequence youd like, ...
 - a. take note of the Accession number along with the amount of base pairs
 - b. Select the FASTA button, then on the page it links you to, copy and past the entire sequence (Starting at the carot (>) until the very end)
 - i. EX: >KF791053.1 *Vaucheria litorea* strain BW3 ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit gene, partial cds; chloroplast
ATTGTAATGATTGATCTTGTAATTGGTTATACTGCTATTCAATCT
ATGTCAATTTGGGCTAGAAAAGCTG
ATATGATTTTACATTTACACCGTGCAGGTAATTCTACATATGCAC
GTCAAAAAAATCATGGTATTAATTT
CCGTGTAATTTGTAAATGGATGCGTATGGCTGGTGTGCGATCAC
ATTCATGCTGGTACTGTTGTAGGAAAA
TTAGAAGGCGATCCTTTAATGGTTAAAGGCTTTTATAATACGTT
ATTATTAATAAATTAGAAATTAATC
TTTCTCAAGGTGTATTTTTTGAATGGATTGGGCAGCATTACGT
AAAAGTGTACAGTAGCATCTGGTGG
TATTCATGTGGACAATTACATCAATTA
 - c. Download that page as a basic text document and save for next steps

Blast Analysis (NCBI Blast, Webversion)

4. Search for *NCBI Blast* on the web
 - a. Should be first link “BLAST: Basic Local Alignment Search Tool” by the *National Institutes of Health (.gov)*
5. Select the option on the left, *Nucleotide Blast*
 - a. It should take you to a new page with a row of boxes that say “blast_” (or something similar) make sure the blue highlighted one is *blastn*.
6. Enter the accession numbers of the FASTA you want to sequence, separate them using a common (,).
7. Name the job
8. Select the *BLAST* option on the bottom
9. It will show a new page, and on that new page, on the bottom, there's a chart with several sequences. Make note of the *Query Cover*, *E-Value*, & *Per. Ident*.
10. Save the results, including the top result from the table at the bottom.

T-Coffee (Webversion; Version_11.00)

11. Search for *t-coffee alignment*. Select the link that says *T-COFFEE Multiple Sequence Alignment Server*

12. Of the three options presented, move your cursor over the one that says *DNA*, and then select *Combine popular aligners (M-Coffee)*
13. On the new page, it asks for your sequences.
 - a. Duplicate the plain text doc used for the NCBI Blast.
 - b. On the new doc, remove everything after the organism name and beginning of the sequence
 - c. Upload the updated plain text doc and wait.
 - i. Optional: add your email to receive a notification when its been completed.
14. Screenshot the alignment results
15. Scroll down to where it says *Result files* and download the *ph file* (used for iTOL) and the *clustalw_aln file*
 - a. If it doesn't download and instead goes to a blank webpage with the content super small, I copy/pasted it into an empty google doc, then downloaded it as a plain text file. You can also add the entire species name rather than just the genus on this text file.

N•ESPrpt 0.9 (Webversion, NESPrpt.cgi 4.1.5)

16. Search *NESPrpt*, and do NOT select the option that says *ESPrpt 3*, instead look for the one that says *N•ESPrpt*.
17. Select the green option up-top that says *RUN N•ESPrpt*
18. On the new page, make sure it's set to *DNA* and not *RNA*
19. For the input file, add the *clustalw_aln* file from T-Coffee.
 - a. If it gives you an error, you may have to delete the stuff from top of the plain text doc (everything before the organism names)
20. Optional: Under *Sequence similarities depiction parameters*, check the box that says to *Display consensus seq.*
 - a. Makes it easier to see where a mutation occurred, saves time.
21. Select the yellow *SUBMIT* option up-top.
22. It will open a new page and on that new page, there is a results table. On the results table, select the option that is BELOW *PostScript file*, but to the right of *PDF file*. If you move your cursor over it, it should highlight orange and be underlined.
 - a. For me, I would select the option that says *pdf [17 Kb]*. Yours may say something similar.
23. Save this pdf output.

Phylogenetic Analysis Using the DNA Alignment on iTOL

24. Search *itol*, and select the option that says *iTOL: Interactive Tree of Life*
25. In the top of the homepage, there's the *Upload* option, select that
 - a. If working in split-screen, there's three horizontal bars. Select those, then select the *Upload* option.
26. Once you're brought to the new page, name your tree and upload the *ph file* file you saved back in step 15 (T-Coffee)

27. Once your tree has been created, go to the advanced setting, then Branch metadata display, and set both *Node IDs* and *Branch lengths* to *Display*. This will turn on your bootstraps.
28. Finally, mess around with the control panel and make it your own!

Notes to Keep in Mind:

- Smaller amounts of basepairs run faster than larger ones and it helps to keep the amount of basepairs relatively similar to one-another